

LOGESH S

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EXECUTIVE SUMMARY

AI & Deep Learning Engineer with strong hands-on experience designing and deploying end-to-end AI solutions across Generative AI, LLMs, RAG, Computer Vision, and Predictive Modeling. Proven ability to develop research-driven deep learning architectures, production-ready AI applications, and scalable intelligent systems, translating complex AI concepts into practical business solutions. Proficient in PyTorch, TensorFlow, LangChain, FastAPI, Docker, and modern AI engineering workflows, with experience spanning model development, system integration, optimization, and deployment.

EDUCATION

- **B.Tech — Artificial Intelligence and Data Science** 2022 – May 2026
Kangeyam Institute of Technology, Tirupur, Tamil Nadu CGPA: 8.53
Relevant Coursework: Data Science, Machine Learning, Deep Learning, Python Programming, Data Visualization
- **HSC — Navarasam Matric Hr Sec School, Erode, Tamil Nadu** 2021 – 2022 | Percentage: 77.8%

SKILLS

- **Languages:** Python, SQL, Statistics
- **GenAI & LLM Stack:** LangChain, LangGraph, LlamaIndex, RAG Pipelines, Hybrid Retrieval, Prompt Engineering, Embeddings & Vector Search (FAISS, ChromaDB), Groq API, OpenAI API, On-device LLMs (GGUF)
- **AI/ML Stack:** PyTorch, TensorFlow, Vision Mamba, TabNet, KAN (Kolmogorov–Arnold Networks), Computer Vision, NLP, Test-Time Adaptation (TTA), Deep Learning Architecture
- **Backend MLOps:** FastAPI, Flask, REST APIs, Docker, docker-compose, MLflow, PostgreSQL, Model Deployment

PROJECTS

Hybrid RAG AI Assistant *Python, FastAPI, Groq API, FAISS, MLflow, Docker*

- Built a production RAG system using FastAPI, FAISS vector search, and Groq (Llama 3.3 70B) achieving sub-800ms response latency.
- Implemented hybrid retrieval combining dense embeddings with live web search, eliminating hallucinations for 1,000+ document chunks.
- Containerized stack with Docker and integrated MLflow for tracking latency, token usage, and quality metrics.

LEA-TTA — Robust Lane Detection Framework *Python, PyTorch, Test-Time Adaptation, Computer Vision*

- Designed a Lane Existence-Aware Test-Time Adaptation (LEA-TTA) framework for robust lane detection on unstructured roads.
- Engineered core modules including LEAM (Lane Existence Score), ARC (adaptation regulator), and geometry-aware regularization (GCAM).
- Pre-trained on CULane and evaluated adaptation efficiency across IDD and BDD100K datasets, optimizing Plain IoU and F1 metrics.

BioMamba-FiLM — Multimodal Brain Tumor Detection *Python, PyTorch, Vision Mamba, TabNet, KAN*

- Engineered a multimodal medical imaging pipeline combining Vision Mamba for MRI feature extraction and TabNet for clinical/genomic learning.

- Applied Feature-wise Linear Modulation (FiLM) to guide MRI representations using biological vectors for precise glioma grading.

Multi-User Real-Time Eye Gaze Tracking

Python, OpenCV, MediaPipe, Tkinter

- Engineered a real-time gaze tracking system using PyTorch and OpenCV, achieving stable 28+ FPS across 5 simultaneous users.
- Improved gaze accuracy by fusing iris tracking, head-pose estimation, and regression models to reduce tracking drift.

AI-Powered LinkedIn Automation Bot

Python, Google Gemini 2.5 Flash, OAuth 2.0

- Built an LLM-powered content pipeline using Google Gemini API that generated and published posts, reducing manual effort by ~80%.
- Implemented secure OAuth 2.0 authentication for programmatic posting via the LinkedIn API.

EXPERIENCE & INTERNSHIPS

Freelance Deep Learning Engineer | ResearchBrains

May 2026 – Present | Erode, India

- Developed advanced lane detection and adaptation frameworks, optimizing Plain IoU, F1 scores, Expected Calibration Error (ECE), and per-frame latency.
- Delivered rigorous experimental validation reports and modular research pipelines for unstructured road datasets.

AI Intern | LongArc

Jun 2026 – Sep 2026 | Bangalore, India

- Engineered AI-powered supply chain optimization tools and an AI Warehouse Layout Optimization system using PyTorch, improving space utilization by 18%.
- Deployed real-time inference pipelines for warehouse diagnostics, reducing manual route calculation time by 30%.

AI Intern | Infosys Springboard

Nov 2025 – Jan 2026 | Remote

- Built EcoPackAI — a sustainable packaging recommendation platform using Flask, PostgreSQL, and XGBoost predictive modeling.
- Developed a full-stack BI dashboard to visualize sustainability metrics, enabling data-driven packaging decisions for enterprise clients.

AI Intern | Lavendel Consulting

May 2025 – Aug 2025 | Coimbatore, India

- Delivered a production multi-user eye gaze tracking system using MediaPipe and OpenCV, meeting performance targets for 5 concurrent users.
- Handled full pipeline including face recognition, per-user calibration, and real-time model rendering.

LEADERSHIP EXTRACURRICULARS

- **President — Association of AI & Data Science** | Kangeyam Institute of Technology Aug 2025 – May 2026
Led a 100+ member student association — organized technical events, hackathons, and industry speaker sessions; drove corporate collaborations for student exposure to real-world AI roles.
- **Secretary / Mentor — Rotaract Club** | Kangeyam Institute of Technology Jul 2024 – May 2026
Managed 25+ events as Secretary; transitioned to Mentor role to guide juniors in technical projects and leadership development.

CERTIFICATIONS

Joy of Computing Using Python (NPTEL, Oct 2023)

Foundation of Cloud, IoT & Edge ML (NPTEL, Apr 2024)

Artificial Intelligence Primer (Infosys, Dec 2025)

Prompt Engineering (Infosys, Jan 2026)